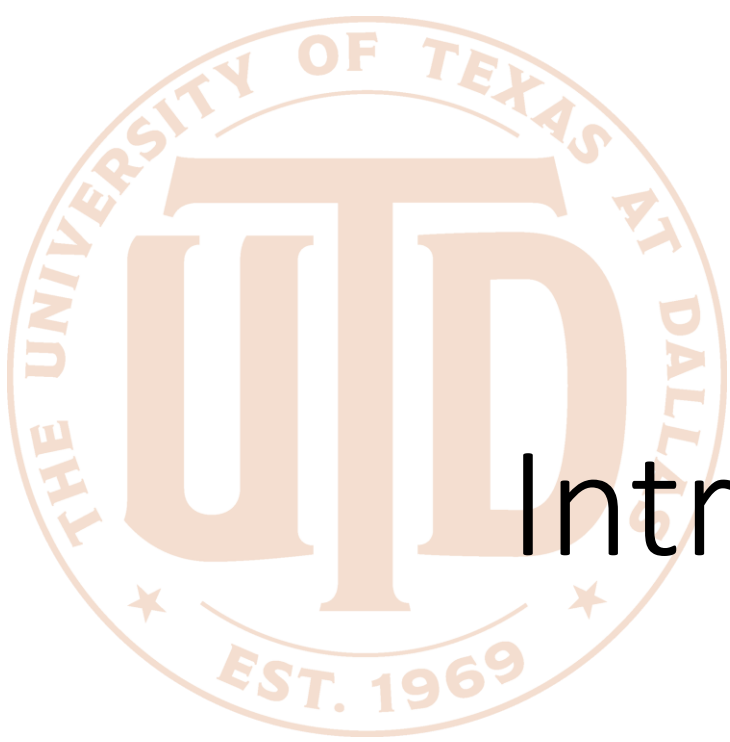




Introduction to Robotics

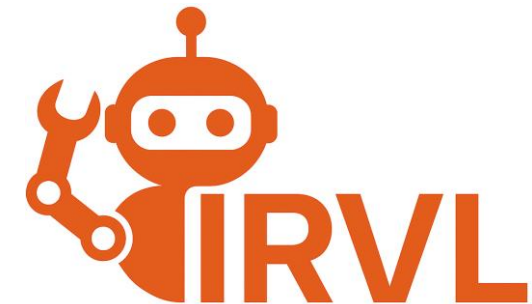
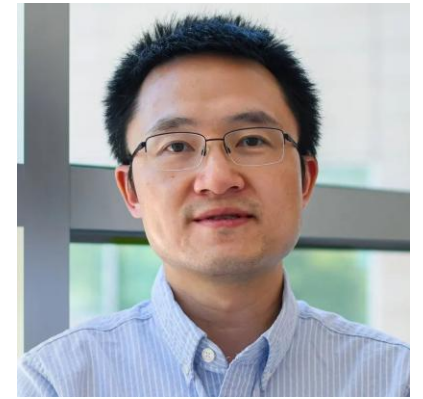
CS 6341 Robotics

Instructor: Yu Xiang

The University of Texas at Dallas

Who am I?

- Assistant Professor in CS at UT Dallas (joined Fall 2021)
- Research area: robotics and computer vision
- Intelligent Robotics and Vision Lab (IRVL)
<https://labs.utdallas.edu/irvl/>



Introduce yourself

- Name
- Major program
- Which year in the program?
- Why are you interested in robotics?



Robots in Factories and Warehouses



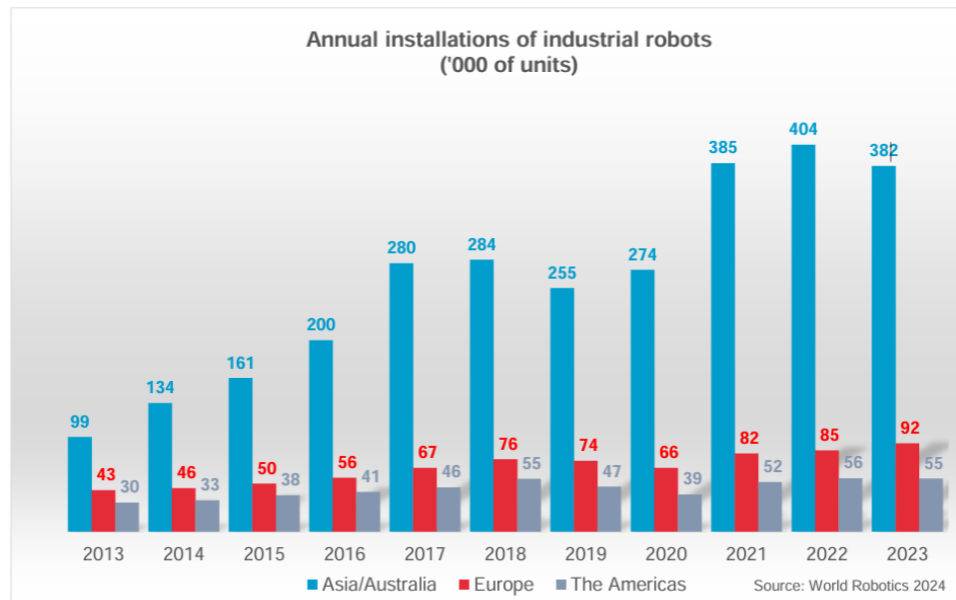
Welding and Assembling



Material Handling



Delivering



<https://ifr.org/wr-industrial-robots/>

Robots in Human Environments



Cleaning Robots



Telepresence Robots



Smart Speakers

How can we have more powerful robots assisting people at homes or offices?

- Mobile manipulators
- Humanoids



Amazon Astro



<https://www.youtube.com/watch?v=YiHE5y1L2Bk>

2022

Google Everyday Robots



<https://www.youtube.com/watch?v=tZn3ekmLvZQ>

2022

Tesla Bot



<https://www.youtube.com/watch?v=XiQkeWOFwmk>

2023

Figure + OpenAI



<https://www.youtube.com/watch?v=Sq1QZB5baNw>

2024

Boston Dynamics + Toyota Research Institute



<https://www.youtube.com/watch?v=HYwekersccY>

2025

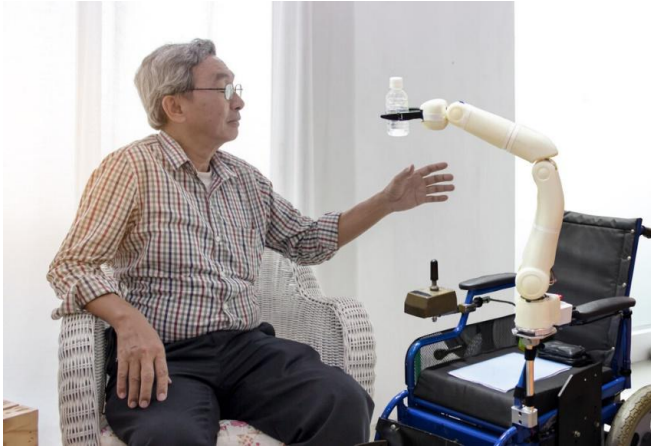
Unitree



<https://www.youtube.com/watch?v=mSPxRVTJW1I>

2025

Future Intelligent Robots in Human Environments



Senior Care



Assisting



Serving



Cooking



Cleaning

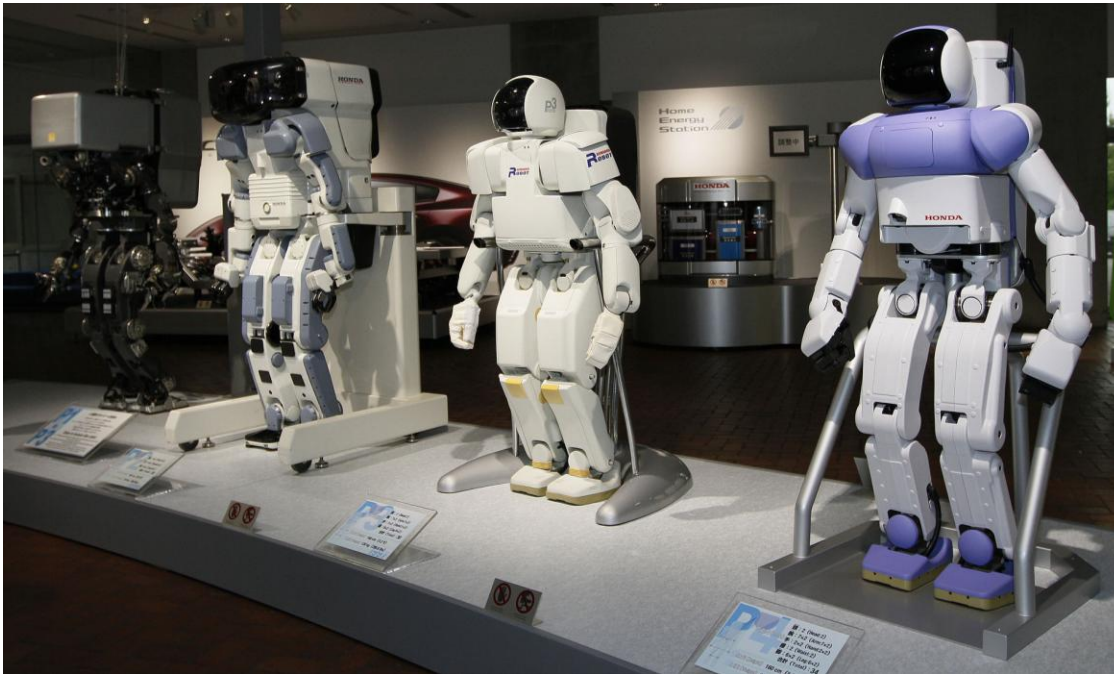


Dish washing

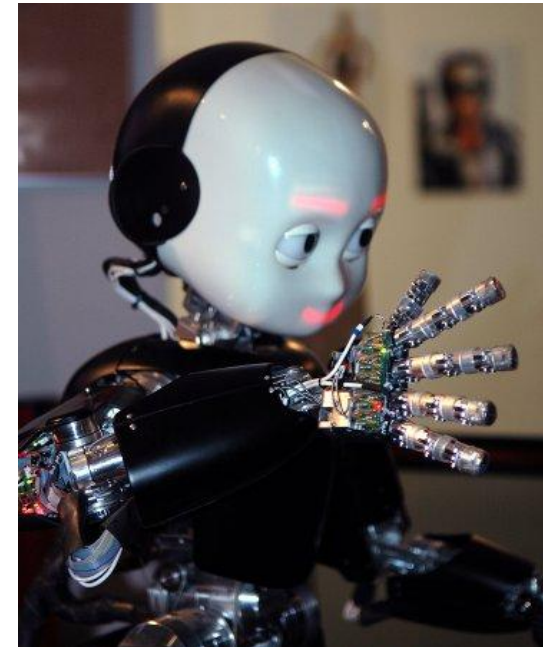
Robot Types

Humanoid Robots

- A humanoid robot is a robot with its body shape built to resemble the human body

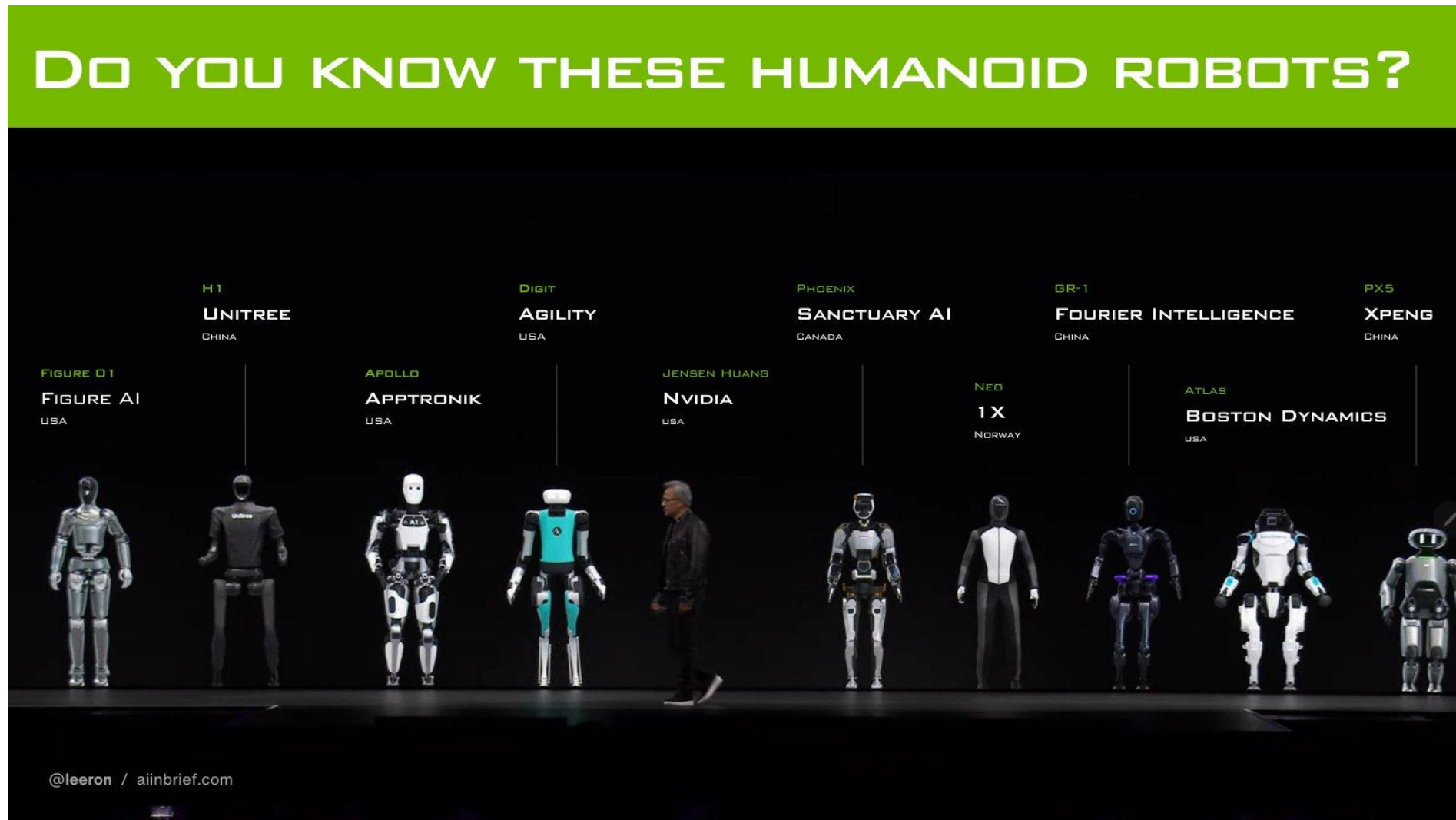


[Honda P series](#)



[iCub](#) robot

Humanoid Robots



Boston Dynamics Atlas



Robot Manipulators

- A device used to manipulate materials without direct physical contact of the operator



ABB



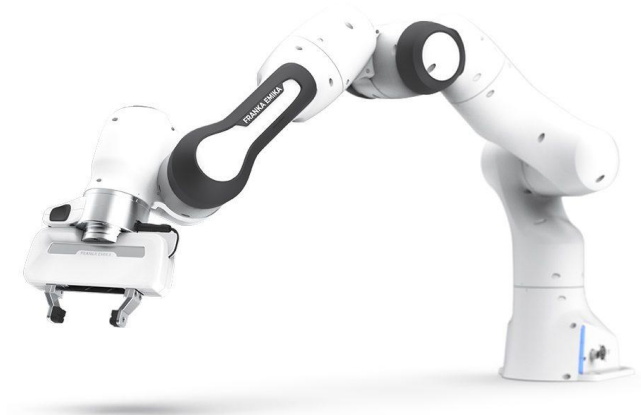
KUKA



FANUC



Yaskawa



Franka Emika

Wheeled Robots

- Use wheels for locomotion
 - Self-driving cars



Starship Technologies



Amazon Astro Robot



Perseverance Rover

Walking Robots

- Legged robots, use articulated limbs to provide locomotion



Boston Dynamics



Robot Cassie

Boston Dynamics



Other Robots

- Flying robots
 - Drones
- Swimming robots
 - Underwater gliders
- Snake robots



Robotic Fish: *iSplash-II*



Two robot snakes. Left one has 64 motors (with 2 degrees of freedom per segment), the right one 10.

Robots vs. Humans

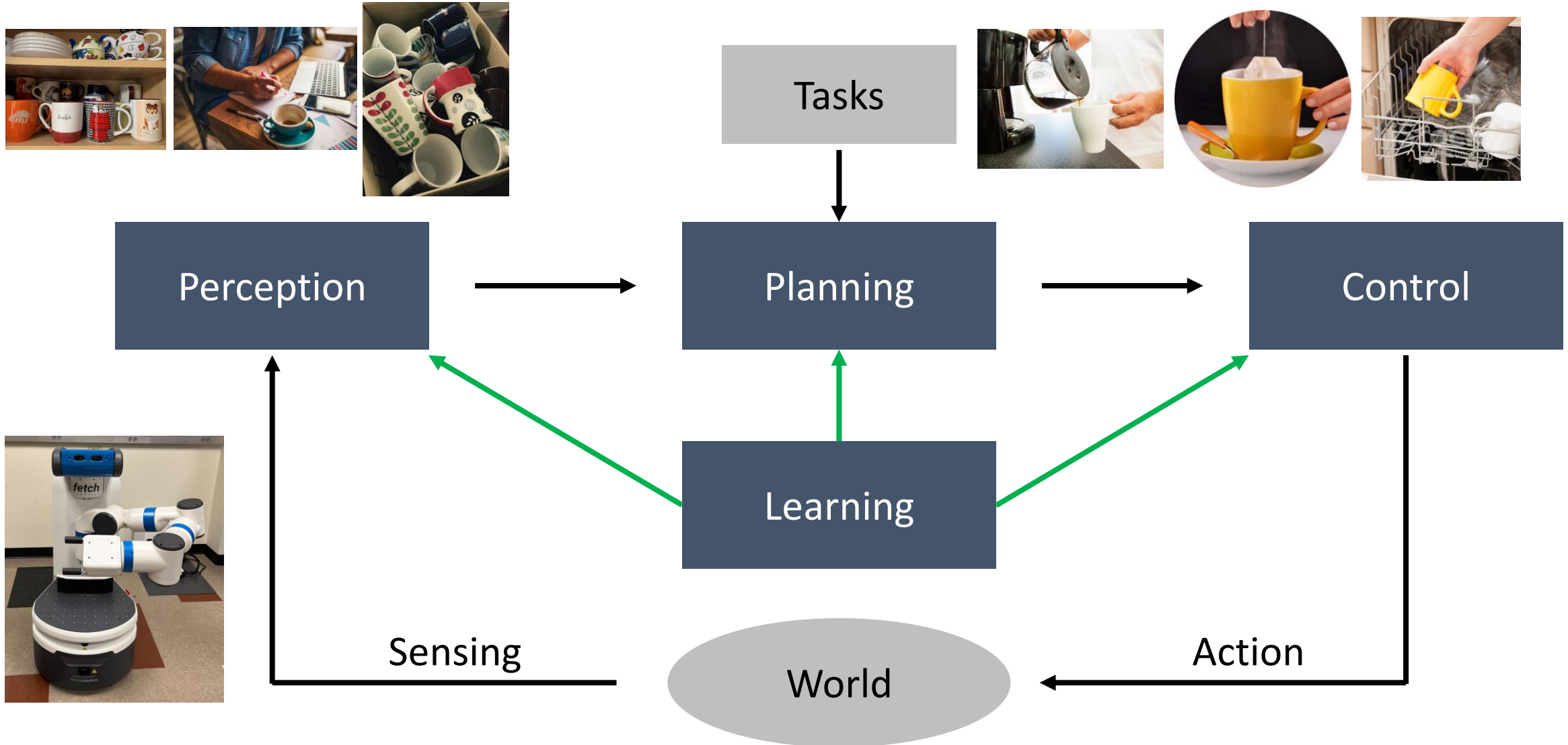
- Sensing
 - Robots: cameras, Inertial Measurement Units (IMUs), joint encoders
 - Humans: vision, vestibular, proprioceptive senses
- Control
 - Robots: motors
 - Humans: muscles
- Computation
 - Robots: robot brain, AI?
 - Humans: human brain

What is a Robot?

What is a Robot?

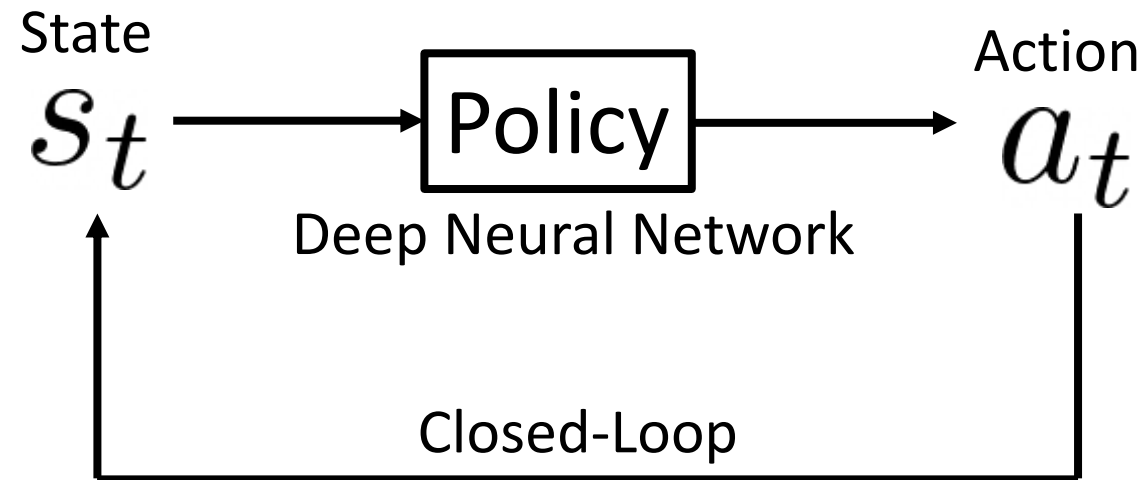
- A robot is a machine capable of carrying out a complex series of actions automatically (Wikipedia)
- A goal-oriented machine that can **sense, plan** and **act**
 - A robot senses its environment and uses that information, together with a goal, to plan some action
 - The action might be to move the tool of an arm-robot to grasp an object, or it might be to drive a mobile robot to some place

Robotic Systems



Robotic Systems: Learning Perspective

- Vision-Language-Action Models
- Video-Action-Models
- World-Action-Models
- Etc.



Robot Applications

- Robot Manipulation
- Robot Navigation

Robot Manipulation

- The ways robots interact with objects
- Examples
 - Grasping an object
 - Placing an object
 - Pushing an object
 - Opening a door
 - Folding laundry
 - Etc.



https://am.is.mpg.de/research_projects/autonomous-robotic-manipulation

Robot Manipulation

Perception

Robust and Accurate

Planning

High degree of freedom
Multi-modal grasping

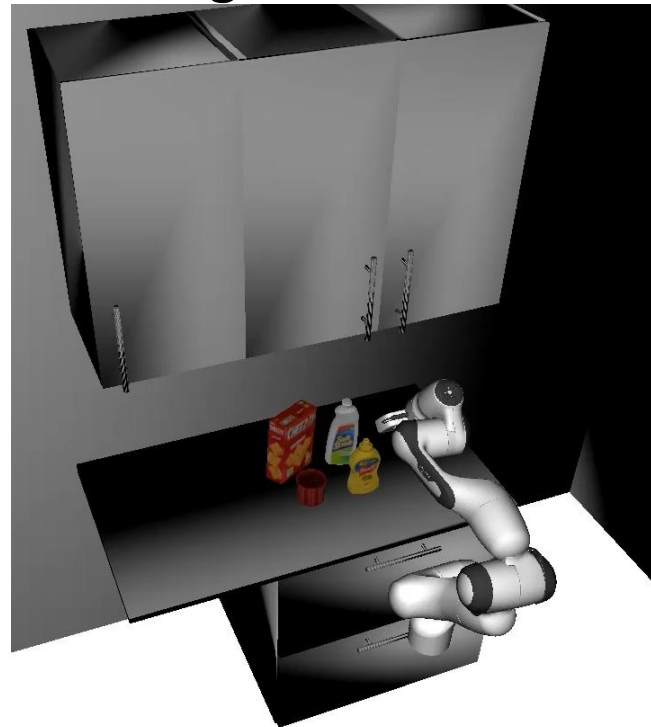
Control

Contact with objects

Sensed image



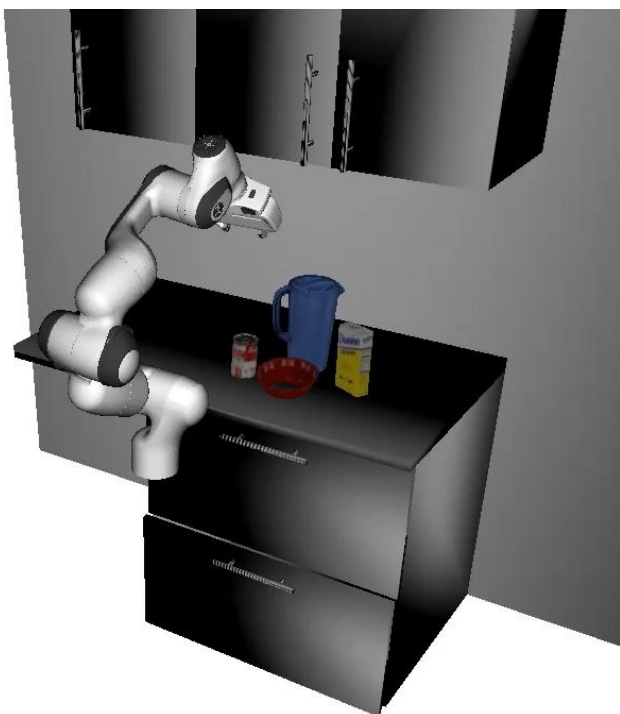
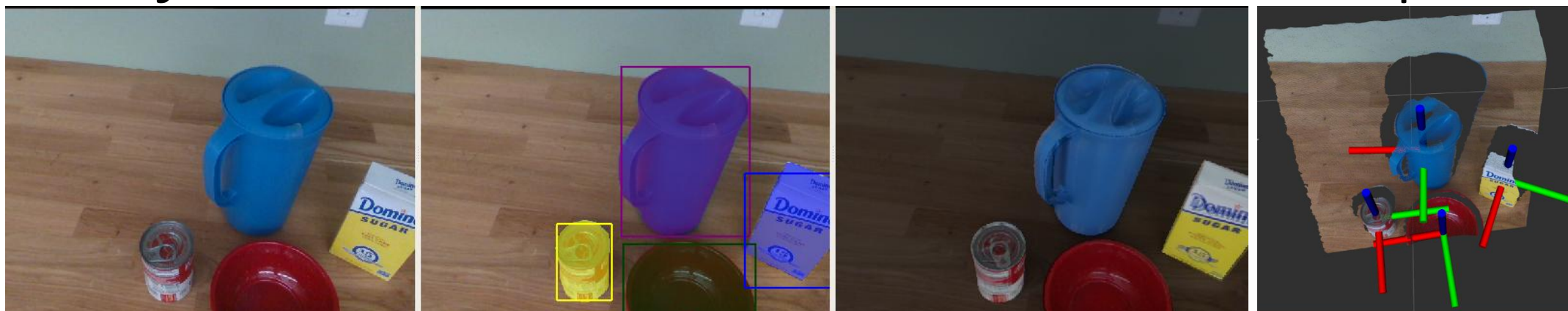
Planning scene



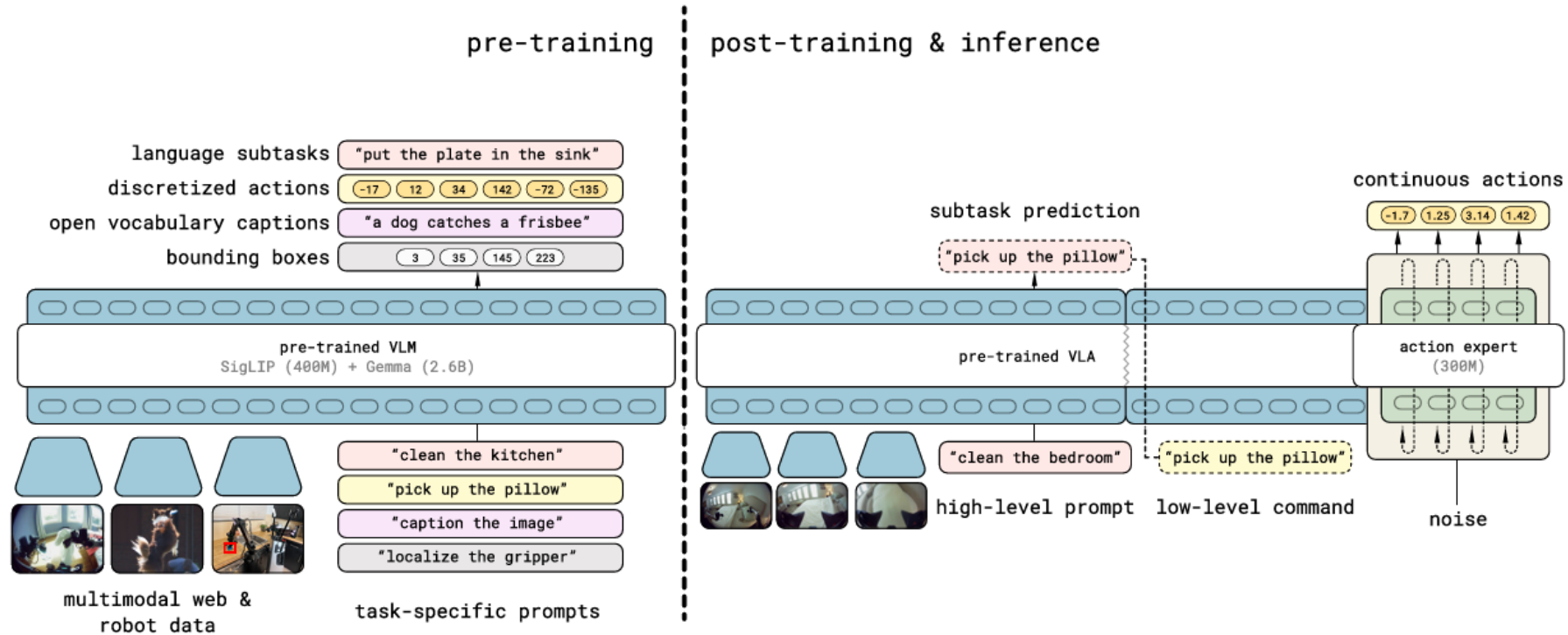
Real world execution



6D Object Pose Estimation for Robot Manipulation



Vision-Language-Action Models



<https://www.pi.website/blog/pi05>

- <https://irvlutd.github.io/VLAReplica/scene-videos/id-pi-0-5.html>

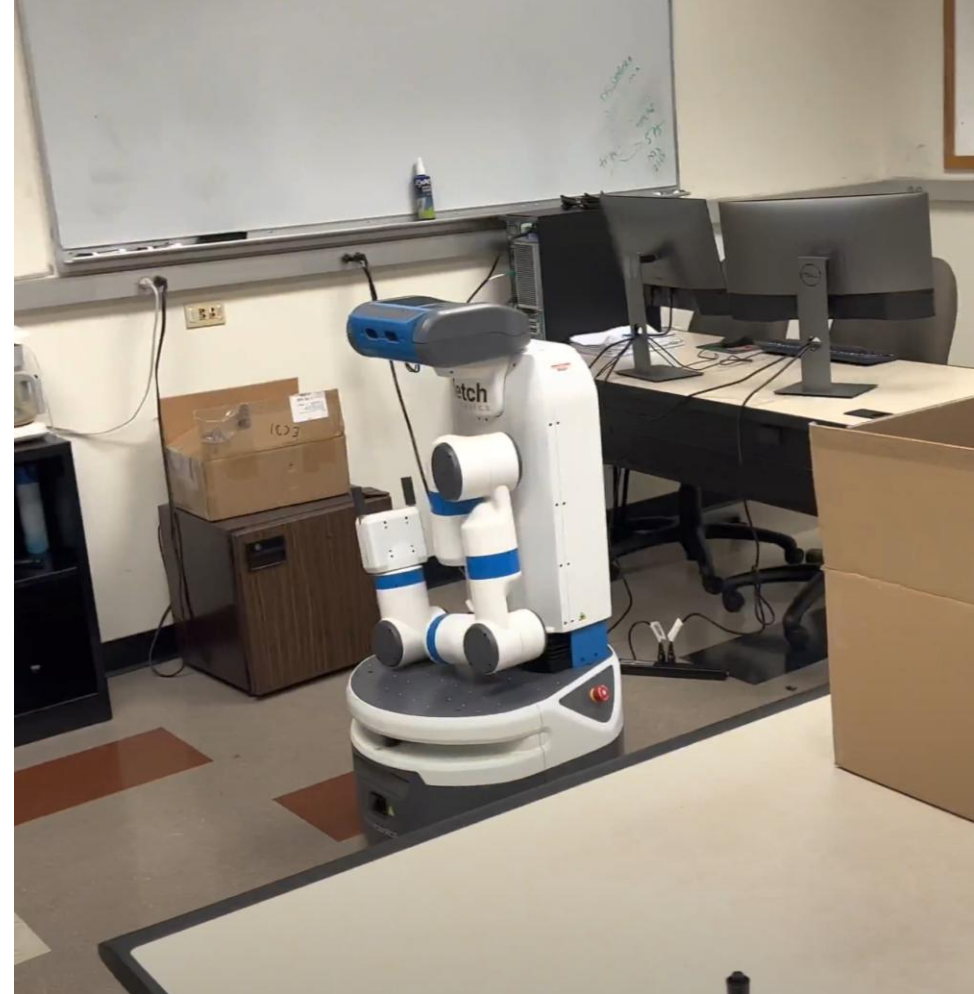
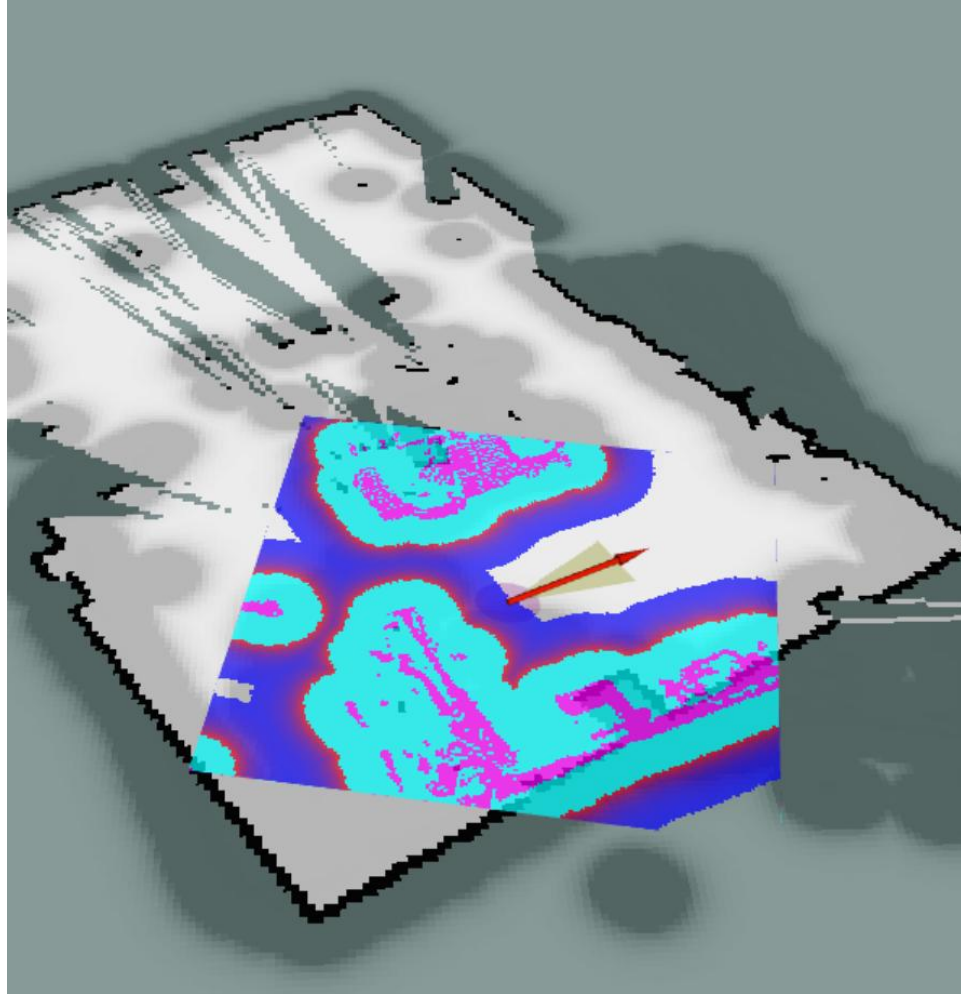
Robot Navigation

- Go from A to B without hitting anything



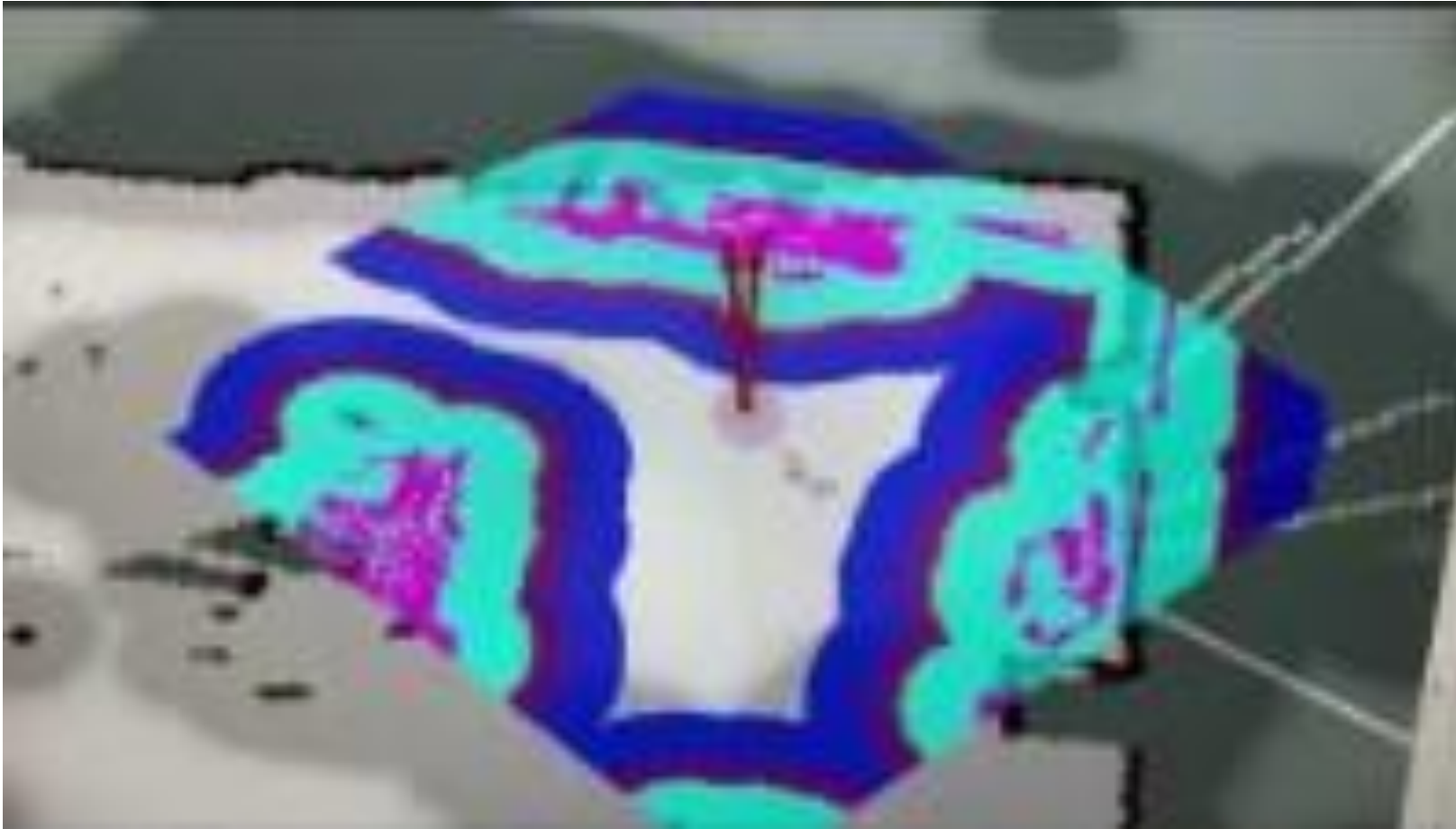
Laser-based SLAM
2D occupancy grid map

Occupancy Grid Mapping



Navigation Demo using ROS

Credit: Gagan Bhat



What will you learn in this course?

- Robot mechanism
 - Design of robot manipulators and wheeled robots
 - Kinematics and dynamics of robots
 - Robot motion planning and control
 - Robot learning
 - Imitation learning, reinforcement learning
 - Robot Operating System (ROS) and robot simulators
- We focus on basics in robotics,
not just robot learning

What will you learn in this course?

- Mathematics in robotics
 - Lectures
- Programming in robotics
 - Homework and projects



Last Year Example



Grading Policy

- Homework (50%)
 - 5 homework in total
 - Individual submission
- Team Project (45%)
 - 3 or 4 students for a project
 - Project proposal (5%)
 - Project mid-term report (10%)
 - Project presentation (15%)
 - Project final report (15%)
- In-class Activity (5%)
 - Quiz
- No final exam

Start thinking about the course project

Course Details

- Textbook
 - Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition <http://hades.mech.northwestern.edu/images/7/7f/MR.pdf>
 - Kevin Lynch's lectures <https://modernrobotics.northwestern.edu/nu-gm-book-resource/foundations-of-robot-motion/>
- My office hour

Tuesday & Thursday 4:00PM – 5:00 PM
ECSS 4.702
- TA office hour: TBD
- Course access and navigation: [eLearning](#)

Questions?